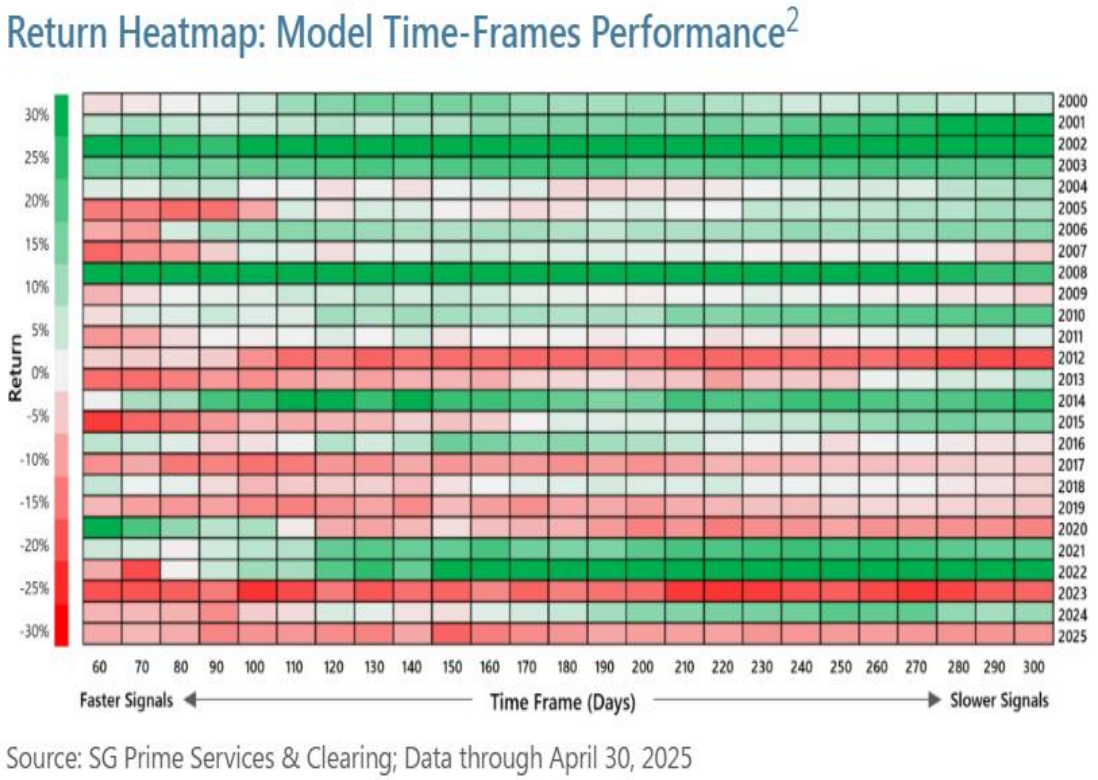
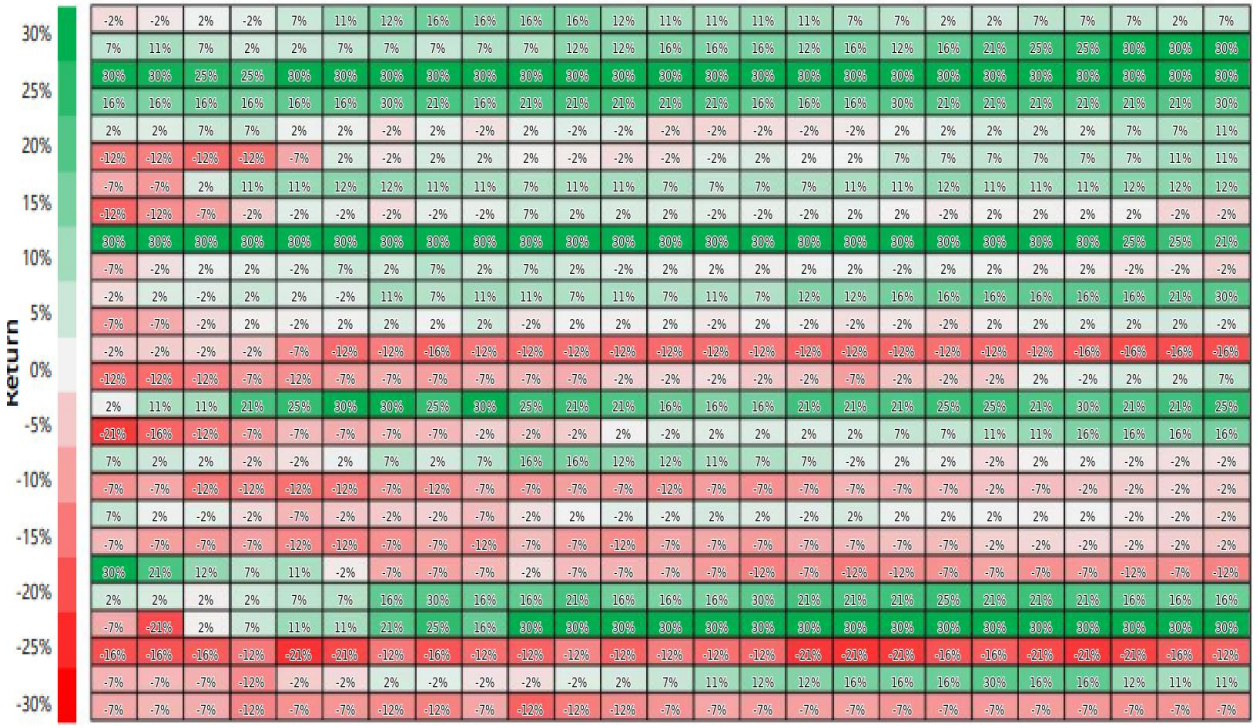
Trend Model Speed Comparison

The heatmap from Graham Capital's *Trend-Following Primer* Research Paper



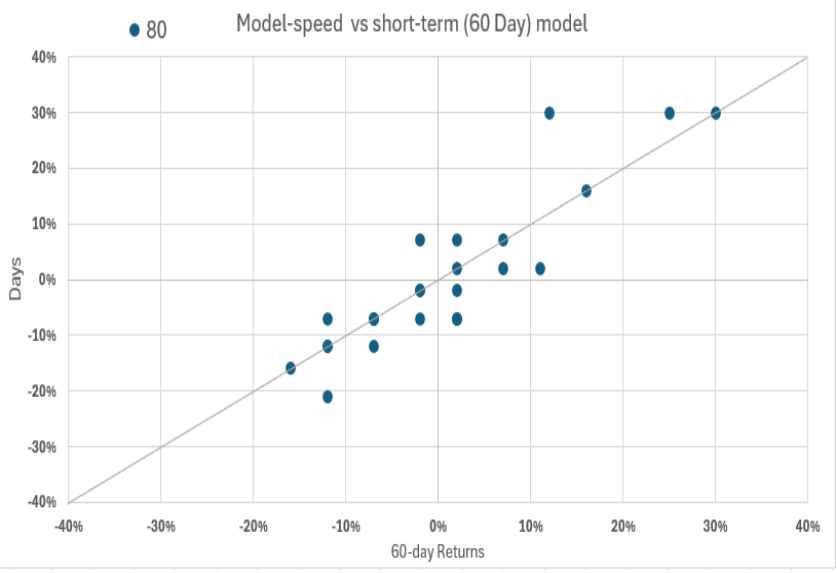
I have used the colour-rank and pixel colours from the heatmap to populate the grid numerically to obtain the data

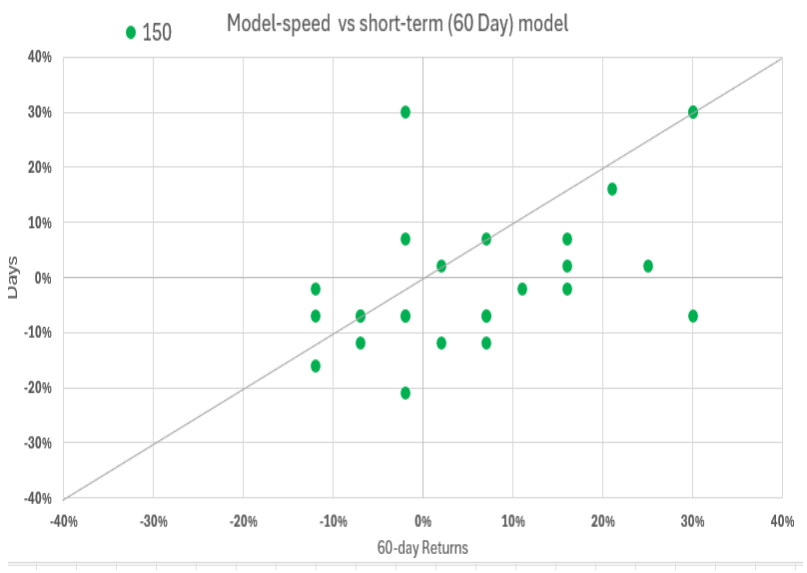


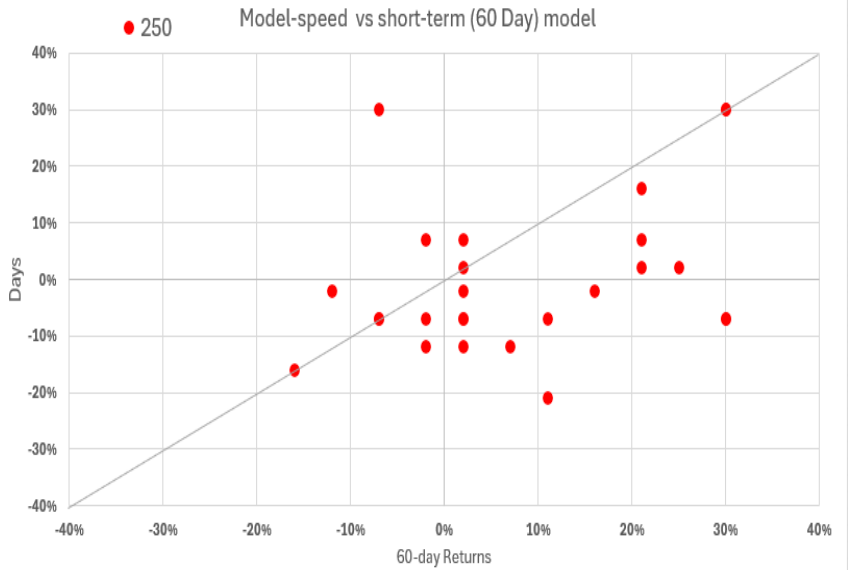
Trend Model Speed Comparison

With the data obtained, I then plotted the returns of different model-speeds (the columns of the heatmap) against a baseline shortest speed of 60 days to see how they compare. I chose a short period (80-day), medium (150-day) and a longer term (250-day).

As expected, the 80-day model is close to the 60-day baseline. As the period increases, the downside improves relative to the 60-day, however it loses some upside. Generally, the long-term models still demonstrate some positive convexity relative to the short-term models.







Trend Model Speed Comparison

I plotted each speed model's annual return against the S&P return to examine convexity of each model.

Convexity: the coefficient of the square term in a quadratic line-of-best fit.

Left graph: 60-day model against the S&P, corresponding to a convexity of 1.3.

Right graph: The convexity measure against the model period to examine which model has the greatest convexity against the S&P.

